

DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology

Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha

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Research Interests

I study **motion** as a computational medium across HCI and computer graphics. My research builds systems for designing, generating, and fabricating motion. I aim to make motion programmable and tangible across animated characters, mechanical artifacts, robotic systems, and AI-assisted creative tools.

Before starting my Ph.D., I spent eight years building software systems across AI systems, games, cloud platforms, and social robots.

Education

Georgia Tech, Ph.D. in Computer Science

2022–Present

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE

2020–2022

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE

2012–2020

Experience

Ph.D. Research Intern, Dolby Laboratories

2024–2025

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp.

2023

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc.

2021–2022

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador

2020–2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc.

2019

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio)

2017–2019

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

- 1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *CHI*, Apr. 2026.
- 2 J. Yu, P. Garg, **D. Synn**, and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *CHI*, Apr. 2025.
- 3 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.
- 4 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmaeilzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS*, 2021.